



UAV



GCS

fresh ML-KEM-1024 and
X25519 key pairs

ClientHello

suite UAV ID N_C cell time X25519 key ML-KEM key ML-DSA-87 sig.

verify the signature
(pinned UAV key),
encapsulate, derive
keys, sign

session ID N_S X25519 key ML-KEM ciphertext ML-DSA-87 sig. MAC **ServerHello**

verify the signature
(pinned GCS key),
decapsulate, derive
keys, check the MAC

ClientFinished session ID MAC

check the MAC:
session confirmed



session keys = $\text{HKDF}(ss_{\text{KEM}} \parallel ss_X, \text{salt} = TH)$

application records under AES-256-GCM